

GENAI ENGINEER ROADMAP

60 Days - Every Day Broken into Parts

Portrait concise edition with difficulty + focus

Difficulty	1-4 Easy 5-6.5 Moderate 7-8 Hard 8.5-10 Very Hard
Focus	Concept / Applied / Framework / Backend / Production / Project / Interview
Parts	Each source topic for a day is numbered as Part 1, Part 2, etc.
Important	Difficulty and focus are planning estimates. Topic/part names come from the original roadmap.

Milestone flow

LLM Foundations -> RAG -> Frameworks/Agents -> Backend -> Production -> Enterprise Projects -> Interviews

Roadmap at a Glance

Original 60-day structure. Our live sessions may compress some foundation topics into fewer calendar days.

Sprint / days	Main focus	Avg. difficulty	Milestone
Sprint 0 Day 1	Set up the engineering workflow.	2.0/10	Engineering setup
Sprint 1 Days 2-16	Understand LLM internals from tokens to inference.	7.2/10	LLM mini project + mock
Sprint 2 Days 17-25	Build retrieval and production RAG capability.	7.3/10	Enterprise RAG project
Sprint 3 Days 26-34	Use LLM APIs, frameworks, agents and orchestration.	6.9/10	Multi-Agent project
Sprint 4 Days 35-42	Build production backend services for GenAI systems.	6.8/10	Production backend project
Sprint 5 Days 43-50	Containerize, automate and deploy.	6.6/10	Deployment project
Sprint 6 Days 51-60	Build enterprise projects, debug production, prepare interviews.	8.4/10	Capstone + career

Key project + interview checkpoints

- Day 16 - LLM revision, interview questions, mock interview and mini project.
- Day 25 - Enterprise RAG Chatbot.
- Day 34 - Multi-Agent Assistant.
- Day 42 - Production Backend.
- Day 50 - Deployment Project.
- Days 51-56 - Enterprise project stretch.
- Days 57-58 - Production monitoring + debugging.
- Days 59-60 - System design / GenAI interviews + final capstone + career preparation.

Sprint 0 - Engineering Foundation

Days 1-1 - Engineering setup

Day	Parts	Focus	Difficulty
1	P1. Repository Setup P2. Git Basics P3. GitHub P4. VS Code P5. Engineering Workflow P6. Documentation Structure P7. Learning Roadmap	Foundation	2/10 Easy

Average difficulty for this block: 2.0/10.

Sprint 1A - LLM Foundations

Days 2-8 - Tokens, embeddings and attention foundations

Day	Parts	Focus	Difficulty
2	P1. Introduction to LLMs P2. Traditional Programming vs AI P3. Tokenizer P4. Token IDs P5. Embeddings P6. Context P7. Attention (Concept)	Concept	5/10 Moderate
3	P1. Query P2. Key P3. Value P4. Query-Key Matching P5. Attention Scores P6. Softmax (Concept) P7. Weighted Values P8. Engineering Assessment	Concept	7/10 Hard
4	P1. Why Embeddings aren't enough P2. Query from Embedding P3. Key from Embedding P4. Value from Embedding P5. W_q P6. W_k P7. W_v P8. Learned Transformations	Concept	7/10 Hard
5	P1. Matrix Basics P2. Vectors P3. Matrix Multiplication P4. Vector Transformation P5. $Q = XW_q$ P6. $K = XW_k$ P7. $V = XW_v$	Concept	7/10 Hard
6	P1. Dot Product P2. Similarity P3. $Q \times K^T$ P4. Attention Score Calculation	Concept	7/10 Hard
7	P1. Scaling ($\sqrt{d_k}$) P2. Why Scaling is Needed P3. Numerical Stability	Concept	7.5/10 Hard
8	P1. Softmax (Mathematics) P2. Attention Weights P3. Weighted Sum P4. Attention Equation	Concept	7/10 Hard

Average difficulty for this block: 6.8/10.

Sprint 1B - LLM Foundations

Days 9-16 - Transformer internals, training and inference

Day	Parts	Focus	Difficulty
9	P1. Self-Attention P2. Attention Matrix P3. Information Flow	Concept	8/10 Hard
10	P1. Multi-Head Attention P2. Multiple Heads P3. Concatenation P4. Output Projection	Concept	8/10 Hard
11	P1. Positional Encoding P2. Sin/Cos Encoding P3. Position Information	Concept	6.5/10 Moderate
12	P1. Transformer Architecture P2. Encoder P3. Decoder P4. Residual Connections P5. Layer Normalization	Concept	8/10 Hard
13	P1. GPT Architecture P2. Decoder-only Transformer P3. Masked Attention P4. Next Token Prediction	Concept	8/10 Hard
14	P1. Pretraining P2. Fine-Tuning P3. Transfer Learning P4. RLHF	Concept	7.5/10 Hard
15	P1. Inference P2. Context Window P3. KV Cache P4. Greedy Search P5. Beam Search P6. Temperature P7. Top-K P8. Top-P	Concept	7/10 Hard
16	P1. Sprint 1 Revision P2. LLM Interview Questions P3. Mock Interview P4. LLM Mini Project	Review + Mini Project	7/10 Hard

Average difficulty for this block: 7.5/10.

Sprint 2 - RAG & Retrieval Engineering

Days 17-25 - Retrieval systems and production RAG

Day	Parts	Focus	Difficulty
17	P1. Embedding Models P2. Embedding APIs P3. Similarity Search	Applied	6/10 Moderate
18	P1. Vector Databases P2. FAISS P3. ChromaDB P4. Pinecone	Applied	7/10 Hard
19	P1. Chunking P2. Metadata P3. Indexing	Applied	6.5/10 Moderate
20	P1. Semantic Search P2. Hybrid Search P3. BM25	Applied	7.5/10 Hard
21	P1. Retrieval P2. Top-K Retrieval P3. Context Building	Applied	6.5/10 Moderate
22	P1. RAG Architecture P2. End-to-End Flow	Applied	7/10 Hard
23	P1. Advanced RAG P2. Query Expansion P3. Compression P4. Reranking	Applied	8/10 Hard
24	P1. Production RAG P2. Evaluation P3. Hallucination P4. Guardrails	Production Applied	8.5/10 Very Hard
25	P1. Sprint 2 Project P2. Enterprise RAG Chatbot	Project	8.5/10 Very Hard

Average difficulty for this block: 7.3/10.

Sprint 3 - LLM Frameworks

Days 26-34 - Prompting, APIs, LangChain, LangGraph, MCP and agents

Day	Parts	Focus	Difficulty
26	P1. Prompt Engineering P2. Prompt Patterns	Applied	5.5/10 Moderate
27	P1. OpenAI API P2. Gemini API P3. Claude API	Applied	5.5/10 Moderate
28	P1. LangChain Basics	Framework	6/10 Moderate
29	P1. LangChain Chains P2. Memory P3. Tools	Framework	6.5/10 Moderate
30	P1. LangGraph Basics	Framework	7/10 Hard
31	P1. MCP (Model Context Protocol)	Framework	7.5/10 Hard
32	P1. AI Agents P2. Agent Loop	Framework	7.5/10 Hard
33	P1. Multi-Agent Systems	Framework	8/10 Hard
34	P1. Sprint 3 Project P2. Multi-Agent Assistant	Project	8.5/10 Very Hard

Average difficulty for this block: 6.9/10.

Sprint 4 - Backend Engineering

Days 35-42 - Backend services for GenAI systems

Day	Parts	Focus	Difficulty
35	P1. FastAPI Basics	Backend	5.5/10 Moderate
36	P1. REST APIs P2. Dependency Injection	Backend	6/10 Moderate
37	P1. PostgreSQL	Backend	6.5/10 Moderate
38	P1. SQLAlchemy P2. Alembic	Backend	7/10 Hard
39	P1. Authentication P2. JWT P3. OAuth	Backend	7.5/10 Hard
40	P1. Redis P2. Background Tasks	Backend	7/10 Hard
41	P1. Logging P2. Monitoring P3. Error Handling	Backend	7/10 Hard
42	P1. Sprint 4 Project P2. Production Backend	Project	8/10 Hard

Average difficulty for this block: 6.8/10.

Sprint 5 - Production Engineering

Days 43-50 - Containers, CI/CD, cloud and deployment

Day	Parts	Focus	Difficulty
43	P1. Docker	Production	5.5/10 Moderate
44	P1. Docker Compose	Production	6/10 Moderate
45	P1. Git Workflow P2. Branching P3. Pull Requests P4. Code Reviews	Production	6/10 Moderate
46	P1. CI/CD	Production	7/10 Hard
47	P1. Kubernetes Basics	Production	7.5/10 Hard
48	P1. AWS Basics	Production	6.5/10 Moderate
49	P1. Azure Basics	Production	6.5/10 Moderate
50	P1. Deployment Project	Project	8/10 Hard

Average difficulty for this block: 6.6/10.

Sprint 6 - Enterprise Projects & Career

Days 51-60 - Portfolio projects, production operations and interviews

Day	Parts	Focus	Difficulty
51	P1. Enterprise RAG Chatbot	Enterprise Project	8.5/10 Very Hard
52	P1. AI Resume Screening System	Enterprise Project	8/10 Hard
53	P1. AI Customer Support Agent	Enterprise Project	8.5/10 Very Hard
54	P1. AI Meeting Assistant	Enterprise Project	8/10 Hard
55	P1. AI Research Agent	Enterprise Project	8/10 Hard
56	P1. Multi-Agent Enterprise Platform	Enterprise Project	9/10 Very Hard
57	P1. Production Monitoring	Production Ops	8/10 Hard
58	P1. Debugging Production Issues	Production Ops	9/10 Very Hard
59	P1. System Design Interview P2. GenAI Interview	Interview	8.5/10 Very Hard
60	P1. Final Capstone Project P2. Resume Review P3. Portfolio Review P4. Mock Interview P5. Job Application Strategy	Capstone + Career	9/10 Very Hard

Average difficulty for this block: 8.4/10.

Difficulty Curve & How the Work Changes

Days	Dominant mode	What becomes hard	Difficulty
Days 1-16	Concept-heavy	Understand internals and first principles.	7-8/10 typical
Days 17-34	Applied GenAI	New concepts + more implementation through RAG and agents.	6-8.5/10
Days 35-50	Engineering-heavy	APIs, databases, reliability, CI/CD, deployment.	5.5-8/10
Days 51-60	Project / interview-heavy	Design, debugging, production judgment and explanation.	8-9/10

Early: harder to understand. Middle: harder to build correctly. Final: harder to design, debug and defend decisions.

Guiding principle

Do not learn technologies in isolation. Learn to solve engineering problems.